

Large-Scale Multimodal Content Analysis and Annotation with Vision-Language Models

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Abstract

Most social media research still treats content as text, even though images and videos carry a large share of what people actually see and share. This gap is especially serious for WhatsApp in India, where political communication often travels as posters, memes, screenshots, short clips, and forwarded videos in Hindi and other low-resource languages. Text-only analysis can therefore miss a large fraction of political content and can distort conclusions about what topics spread and what becomes viral.

We present a large-scale multimodal analysis of WhatsApp data collected via data donation across roughly 100 locations in India during the 2024 Indian General Elections. Using recent vision-language models (VLMs) and large language models (LLMs), we build a multimodal processing toolkit that represents text, images, and videos in a shared framework. We use this toolkit to produce three results. First, we map topic prevalence and diffusion across modalities and show that topic distributions differ sharply by modality: key domains such as politics are disproportionately non-textual, so text-only pipelines systematically undercount them. Second, we evaluate whether LLM/VLM-based classifiers can replace human annotation for socially sensitive labels. Models perform well for broad categories such as political and news content, but they are substantially less reliable for misinformation, hate speech, and caste-coded hostility, with failures driven by class imbalance, implicit meaning, and cultural context. Third, we connect WhatsApp content to mainstream narratives by building a reference set from YouTube uploads of prime-time television news over the same period and measuring multimodal narrative overlap under format transformation (e.g., headlines as screenshots and segments as clips). The narratives that align with mainstream coverage are carried largely by non-text WhatsApp messages and are far more likely to be viral than baseline content.

Together, these findings show that multimodal methods are necessary for valid measurement of political communication on encrypted platforms, and they provide an auditable toolkit for studying diffusion, harms, and cross-channel narrative dynamics in low-resource settings.

1 Introduction

WhatsApp is one of the most important communication infrastructures in India. It is used for everyday coordination,

but it is also a primary route through which political messages circulate, often with high credibility because they arrive through trusted social ties (Trauthig and Woolley 2025). A large research literature has documented WhatsApps relevance for political communication, including the spread of misinformation and hate speech (Narayanan et al. 2019; Varanasi, Pal, and Vashistha 2022). At the same time, a basic fact about WhatsApp in India is still easy to underestimate: much of the content that moves at scale is not textual (Garimella and Eckles 2020). It is multimodal media such as posters, memes, screenshots, short clips, and forwarded videos, frequently mixing Hindi and other regional languages with visuals, audio, and symbolic cues. This creates a measurement problem. If we want to understand what people are exposed to and what spreads on WhatsApp, we need methods that can represent and analyze content across modalities and across low-resource languages, rather than treating WhatsApp as a text archive.

The Indian case also makes this measurement problem politically consequential. The ruling Bharatiya Janata Party (BJP) has invested heavily in digital campaigning and has built a professionalized social media operation, often described as an IT cell, with a substantial presence on WhatsApp (Mahapatra and Plagemann 2019). In parallel, multiple reports describe increasing political and corporate influence over mainstream media through ownership concentration, economic pressure, and constraints on editorial independence (International Journalists’ Network 2024). Together, these conditions motivate a widely discussed hypothesis in journalism and political communication: political narratives may be produced and amplified across channels in a tightly coupled way, with mainstream media and WhatsApp reinforcing each other. The empirical difficulty is that cross-channel tracing is easiest when the objects are purely textual. In practice, the most portable objects between TV, YouTube, and WhatsApp are often screenshots and clips. A narrative from a television segment might be shared as a textual message, a screenshot (image) or a short form clip (video). If we cannot measure multimodal content, we cannot credibly study cross-channel narrative coupling in the form it actually takes.

This paper builds a multimodal processing toolkit for WhatsApp and applies it to a large data-donation dataset collected from roughly 100 locations in northern India during

the 2024 general election. The toolkit uses current LLMs and VLMs to put text, images, and videos in a shared representation, which lets us study topic prevalence and diffusion across modalities, evaluate model-based labeling of harms such as misinformation and hate speech, and measure narrative overlap between mainstream television and WhatsApp.

We organize the paper around three main results:

Result 1: Multimodal Topic Prevalence. We use LLM and VLM representations to estimate the prevalence of high-level topics across text, images, and videos, and we examine how topic distributions shift with virality using WhatsApps “forwarded many times” signal. We show that key topics, especially politics, are heavily non-textual, and that text-only pipelines materially misrepresent both what is discussed and what spreads.

Result 2: LLM’s for Multimodal Annotation. We build an annotation pipeline for sensitive categories such as misinformation, hate speech, and caste-coded hostility across modalities. While models perform reasonably on coarse categories like news and political content, they are not reliable substitutes for humans on socially grounded harm labels. We document systematic failure modes and show how performance varies by modality and by category, supported by qualitative case studies.

Result 3: Linking mainstream television narratives to WhatsApp. We assemble a reference set of mainstream narratives from YouTube uploads of prime-time television coverage over the same period as the WhatsApp data, and use embedding-based matching to measure overlap between television and WhatsApp. The cross-channel narratives we can detect are carried mostly by non-textual WhatsApp messages, and they are far more likely to be viral than baseline WhatsApp content. This is hard to see without bringing images and video into the analysis.

Together, these results provide both substantive findings about political communication on WhatsApp during an election period and a reusable toolkit for computational social science. The same approach, using generative models as measurement tools, should transfer to other settings where text, images, and video are mixed and where the meaning of an item often depends on more than one of them.

2 Related Work

WhatsApp as Political Infrastructure in India. WhatsApp has been widely studied as a key channel for political communication in India, in part because it supports trusted, high-reach diffusion through groups and forwarding under end-to-end encryption. Work on party strategy and online mobilization describes how political actors build organized distribution capacity and use WhatsApp as part of election communication infrastructure (Mahapatra and Plagemann 2019; Shih 2023; Woolley and Howard 2018). Methodologically, progress has been enabled by privacy-respecting data access approaches, especially data donation. Garimella and Chauchard introduce a donation-based tool and protocol that make large-scale WhatsApp research feasible without breaking encryption (Garimella and Chauchard 2024). Using WhatsApp data directly, prior work also shows that non-text content is central to political communication and mis-

information. In particular, Garimella and Eckles document the importance of images in political WhatsApp groups and show that a meaningful share of circulated images match known misinformation (Garimella and Eckles 2020). Complementing quantitative work, Banaji et al. study how misinformation and rumor on WhatsApp can connect to offline harm and vigilante violence, emphasizing reception, social context, and the local meaning of messages (Banaji et al. 2019).

Multimodal Analysis and LLMs in Social Media. A large computational literature argues that multimodality is essential for studying modern political content, since meaning in memes, screenshots, and short clips often depends on the interaction between text and visuals. Benchmark work such as Hateful Memes challenge formalizes this point by constructing examples where unimodal systems fail (Kiela et al. 2020). Recent vision-language models and multimodal LLMs make it possible to analyze large mixed-media corpora without task-specific training. Contrastive pretraining (e.g., CLIP) supports retrieval and semantic grouping, and later systems such as BLIP-2 and LLaVA enable captioning and instruction-following analysis with natural language prompts (Radford et al. 2021; Li et al. 2023; Liu et al. 2023). In parallel, work in computational social science has begun to use LLMs for scalable summarization and interpretation of political video content, demonstrating how model-based pipelines can turn video corpora into analyzable text-like objects (Breuer et al. 2025). At the same time, a growing literature cautions that LLM/VLM performance on sensitive categories such as hate speech and misinformation is uneven across languages, cultures, and modalities. Empirical studies document vulnerabilities and prompt sensitivity in hate detection (Roy et al. 2023), and reviews synthesize concerns about bias, evaluation gaps, and domain mismatch (Albladi et al. 2025). Multimodal hate benchmarks that are multilingual and multicultural further emphasize that model behavior can shift substantially across contexts (Bui, Von Der Wense, and Lauscher 2025). These findings motivate the auditing focus in our annotation analysis.

Cross-Platform Narrative Coupling. A third body of work studies how attention and narratives move between mainstream media and online platforms. Classic agenda-setting theory argues that news media shape what issues become salient (McCombs and Shaw 1972). More recent intermedia agenda-setting work examines two-way coupling in hybrid systems, where social media can influence news agendas and legacy outlets can still dominate attention cycles, especially during elections (Harder, Sevenans, and Van Aelst 2017; Guo and Vargo 2020). In India, previous work provides direct evidence of tight coupling between a television debate show and online engagement, consistent with a feedback loop between broadcast agendas and coordinated online activity (Jakesch et al. 2021; Garimella and Datta 2024). This line of work motivates our effort to measure narrative overlap between mainstream media and WhatsApp, but most existing approaches rely on text (transcripts, headlines, hashtags). Our setting requires multimodal matching because television narratives often enter WhatsApp as screenshots and short clips rather than as text.

Political organization and the Indian media environment. Finally, our framing connects to research and reporting on organized digital campaigning and the structure of Indian media ecosystem. Work on computational propaganda emphasizes that influence operations are often hybrid and organizational, combining institutions, labor, and platform affordances (Woolley and Howard 2018). In India, observers have documented the growth of party-linked digital operations and the normalization of coordinated messaging (Mahapatra and Plagemann 2019; Shih 2023). Parallel reporting describes political and corporate pressure on mainstream media and concerns about declining editorial independence (International Journalists’ Network 2024; Reporters Without Borders 2025). These conditions motivate the frequently discussed “central source” idea: that narratives may be pushed through aligned television coverage and WhatsApp dissemination from shared organizational infrastructure. A key barrier to testing such claims is that the most direct traces of coupling are often multimodal artifacts (clips, screenshots, posters). Our work addresses this modal gap by providing tools to link mainstream narratives to WhatsApp circulation even when the carrier is not text.

3 Data Collection

We assemble a multimodal dataset from WhatsApp and mainstream news on YouTube during the 2024 Indian General Elections, which we use to study multimodal topics and virality on WhatsApp, model-based harm annotation, and cross-channel narrative overlap.

3.1 WhatsApp data

Our primary dataset consists of WhatsApp messages collected through a large-scale data donation drive in Uttar Pradesh (UP), India’s most populous state with over 220 million people. We sampled 100 locations across UP, covering urban, semi-urban, and rural settings. Locations were selected to approximate the states demographic composition along key dimensions (age, religion, and caste), benchmarked against census data.

Participants were recruited through in-person visits by our survey team within each sampled town. Following IRB-approved consent procedures, 3,127 participants agreed to donate their WhatsApp data. Data collection used the privacy-preserving tool from (Garimella and Chauchard 2024), which strips sensitive user information at the source. The anonymization and data storage protocols are described in our prior work (anonymized for peer review). For this paper we analyze only the message content (text, images, and video); we do not use any user metadata or demographic information.

Data collection ran for five months, from April to September 2024, covering the full Indian General Election cycle. The raw corpus has over 5.5 million messages from 4,818 unique groups, spanning a wide range of social contexts: work/business, family and friends, religious organizations, local village councils, and news groups. Only about 10% of the groups are explicitly labeled as political, so most of the political content we see is sitting inside non-political groups.

An important feature of the dataset is that WhatsApp itself flags some content as “forwarded many times,” indicating that it has been forwarded at least five hops from the original sender.¹ We use this flag as our measure of viral content. In the full corpus, 55,936 messages (about 1%) carry this flag. Because multimodal analysis is expensive, we construct an analytic dataset that keeps all viral content but downsamples the rest: we include every “forwarded many times” message and add a uniform 5% random sample of the remaining (non-viral) messages. This gives a final analytic dataset of 273,913 messages. Of these 273k messages, 119,589 are text (43.7%), 96,698 are images (35.3%), and 57,626 are videos (21.0%). Images and videos together make up 56.3% of the analytic dataset, which is one reason text-only analysis is not enough in this setting.

3.2 Mainstream News Data (YouTube)

To compare WhatsApp content with mainstream coverage, we collected television news uploaded to YouTube over the same window, April 2024 to September 2024.

Using the YouTube Data API (v3), we pulled video metadata (ID, title, description, tags, view/like counts, and duration) from the official channels of five prime-time news anchors: Aman Chopra, Amish Devgan, Arnab Goswami, Navika Kumar, and Sudhir Chaudhary. These are among the most-watched anchors on Hindi and English prime-time news, and we treat them as a proxy for dominant broadcast narratives.

To group these videos into topics, we processed the titles using the Gemini API in two passes: first we clustered titles week by week into abstract and specific topics, then we manually merged similar clusters so that each topic corresponded to a distinct event or storyline (for example, Lok Sabha election coverage, specific political controversies, or major breaking news). This produced 1,607 headlines organized into 98 topics.

We use this corpus as a reference set of mainstream narratives for the cross-channel matching analysis. It is not “ground truth” about exposure or influence; it is a time-indexed sample of what these five prime-time shows were covering, which we can compare to WhatsApp content at the topic and narrative level.

4 Methods

4.1 Topic detection across modalities

Our main pipeline converts everything to text. For each image and video we generate a dense textual description using a VLM, and then run all downstream analysis on these descriptions together with the original text messages. Examples of generated descriptions are in Section D (Appendix). The practical advantage is that one set of tools (topic models, embeddings, classifiers) works for the whole corpus, regardless of the source modality.

We considered the alternative of embedding all modalities into a shared vector space, e.g. ImageBind or CLIP-style models (Girdhar et al. 2023; Radford et al. 2021). In

¹The exact definition by WhatsApp can be found here <https://faq.whatsapp.com/1053543185312573>

our setting (as of late 2025), these joint embeddings did not reliably align all three modalities, especially for Hindi text mixed with culturally specific imagery, so we did not use them as the primary representation. The text-description approach, in contrast, gave us outputs we could read and audit. We spot-checked the descriptions by hand and found them adequate for capturing both explicit content and most implicit meaning in the messages.

For topic labeling we use Qwen2.5-7B-Instruct on the text descriptions (rather than directly on raw images or video frames). For each message, the model produces two outputs: (1) a primary topic from a fixed taxonomy, and (2) 5–10 secondary keywords summarizing finer-grained content. We run inference in batches with sampling temperature 0.001 to keep outputs as deterministic as possible.

The primary taxonomy has 14 categories chosen to fit the Indian setting: *Religion, Politics, Entertainment, Commerce, Poetry, Employment, Violence, Health, Education, Environment, Greetings, Relationships, Sexual Content, and Miscellaneous*. We arrived at this list iteratively by reading samples of the data and adjusting categories so that they covered both viral content and the day-to-day messages we saw in rural WhatsApp groups.

The prompt templates and the strict output schema enforced for this task are provided in the Appendix C.

4.2 Data annotation

To evaluate whether LLMs and VLMs can be used for automated content analysis, we first built a manually annotated dataset to serve as ground truth. We recruited 40 annotators from the same locations as the data donors so that they could read the local linguistic and cultural cues. Because parts of the corpus contain graphic violence, hate speech, and sexually explicit material, annotators were warned about the nature of the content before they consented and were allowed to skip any item. Each item was annotated by at least three independent annotators, and we used majority vote for the final label. Annotators were trained, and their work was audited periodically by the authors.

Inter-annotator agreement (Krippendorff’s alpha) was in the 0.3–0.5 range for most categories, which is in line with previous work on subjective tasks such as hate speech and political content annotation (Del Vigna12 et al. 2017; Ousidhoum et al. 2019).

We annotated each item along several dimensions that are relevant for studying problematic content in this setting:

1. **False or misleading claims:** We used the broader label “misleading” rather than “misinformation,” so we could include things like cheapfakes or out-of-context media that may be factually real but deceptively framed.
2. **Hateful content:** We annotated all forms of hate speech but paid particular attention to anti-Muslim narratives (“Love Jihad,” “Population Jihad,” “Forced Conversions”), since these accounted for most of the identity-based hostility we saw.
3. **Topical labels:** We also labeled (3) casteist content; (4) health-related advice; (5) news-related information; (6)

partisan political content; and (7) religious content and iconography.

Annotators worked through a custom interface that placed each target item back inside its original WhatsApp group timeline. They could scroll through surrounding messages to see the conversational context before labeling, rather than rating items in isolation.

Out of the 273k items, 16,379 messages were manually annotated and used for the evaluation in Section 5.2. Table 1 shows the breakdown by modality and virality.

For automated annotation we built a separate pipeline that mimics the manual labeling. We wrote prompt templates with definitions, a few-shot examples, and step-by-step instructions. We do not train or fine-tune any model on the annotated data; all model labels come from zero-shot or few-shot prompting. The manually annotated set is used only for evaluation: we compare model labels against the human majority-vote labels and report per-category precision, recall, and F1.

We ran the automated pipeline on all viral content and on the same 5% random sample of non-viral content, choosing a model per modality. Text was handled by Qwen2.5-7B-Instruct; images by Google’s Gemma3-27B. For video we used a tiered approach: viral videos were processed with Gemini 2.5 Pro, and non-viral videos with Qwen2.5-VL-32B-Instruct.²

We constrained all model outputs to predefined JSON schemas using the lmformatenforcer library, so that we could extract several binary and categorical fields per item in a single call. The full prompts and schemas are in Appendix C.³

Automated annotation was run on a cluster of A100 GPUs in bfloat16 precision with a low sampling temperature. For Qwen2.5-VL, video was sampled at 1 frame per second and downsampled to a maximum resolution of 360×420 pixels.

Modality	Non-viral	Viral	Total
Image	4,735	1,212	5,947
Text	7,053	601	7,654
Video	788	1,990	2,778
Total	12,576	3,803	16,379

Table 1: Distribution of manually annotated content by modality and virality.

²At the time of analysis (October 2025), Gemini was the only commercial model that could natively process video with audio, but using it on the full set was too expensive. We therefore used it only for viral videos. For the rest we used Qwen2.5-VL, which uses a native dynamic-resolution ViT with Window Attention and Multimodal Rotary Position Embedding (MRoPE) aligned to absolute time, and which is reported to be competitive with GPT-4o and Claude 3.5 Sonnet on multimodal benchmarks despite processing video frame-by-frame.

³Code, prompts, and additional materials are available at <https://harsha20032020.github.io/whatsapp>.

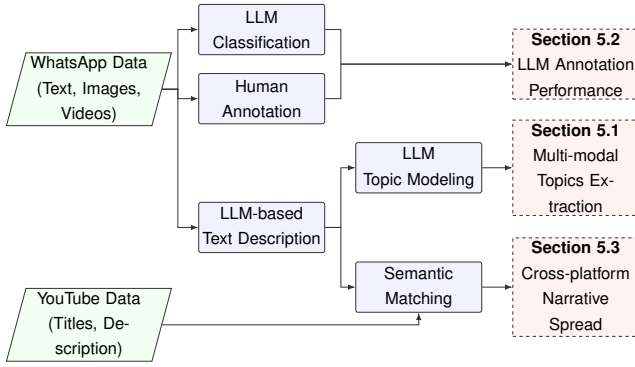


Figure 1: Overview of the methodology pipeline. WhatsApp assets are processed for classification (Sec 5.2) and description generation. Descriptions feed into topic modeling (Sec 5.1) and semantic matching with YouTube data (Sec 5.3).

4.3 Matching YouTube and WhatsApp Content

Linking content from public broadcast media to private messaging is hard because the carrier often changes between platforms (a TV segment becomes a clip; a headline becomes a screenshot). To get around this, we do all matching in a text embedding space. For each WhatsApp message we build a text representation by concatenating the raw text (where available) with the VLM-generated description from the previous step. This way, image and video messages live in the same space as text messages. WhatsApp representations and YouTube headlines are then both encoded with LaBSE (Language-agnostic BERT Sentence Embedding) (Feng et al. 2022). We chose LaBSE because it handles the Hindi–English code-mixing and regional dialects that are common in our data.

We then apply two filters. First, for each YouTube topic, we restrict the search space to WhatsApp messages sent within ± 7 days of the YouTube coverage. Within that window, we retrieve the top- k WhatsApp messages most similar (by cosine similarity) to each headline in the topic. Second, to avoid spurious one-off matches against a single sensational headline, we require multi-headline agreement: a WhatsApp message has to be similar to several headlines from the same topic before we keep it. We aggregate the similarity scores across matched headlines to rank messages.

We also drop any topic with fewer than 10 matched messages. This leaves 45 of the original 98 topics and about 2,884 unique WhatsApp messages. We spot-checked the matched set by hand and found that it bridges the gap between formal news language and the more informal WhatsApp register reasonably well. We also varied the retrieval thresholds, the number of headlines required for agreement, and the keyword filters; the main qualitative findings did not change.

A summary of our methods and datasets is shown in Figure 1.

5 Results

5.1 Topics Across Modalities

We applied the hybrid language–vision pipeline to every message in the corpus, assigning a primary topic from the 14-category taxonomy and a set of secondary keywords. Figure 2 shows the resulting topic distribution.

Two themes stand out. *Commerce* and *Politics* together account for roughly a third of all messages. The volume of commercial content reflects how heavily WhatsApp is used for local trade and small businesses; the volume of political content is striking given that explicitly political groups make up only about 10% of the sample. Political content is therefore largely embedded in non-political spaces.

We also see a non-trivial share of content under *Violence* (3.2%), mostly graphic clips of accidents, local fights, and raw news footage. This kind of unfiltered content is generally absent from mainstream news but circulates widely on WhatsApp. See Table 3 (Appendix) for definitions and examples of each category.

The remaining categories cover the kinds of content typical of WhatsApp use more generally: *Religion*, *Greetings*, *Relationships*, and *Health*. Since the data come from a random subsample of a demographically balanced donor pool, we read these proportions as a reasonable snapshot of WhatsApp use in Uttar Pradesh.

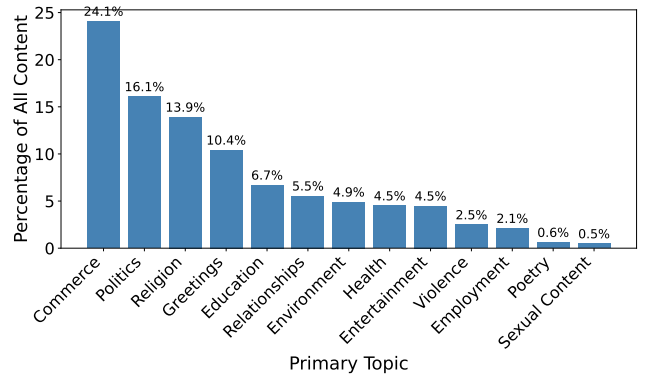


Figure 2: Primary topics as a percentage of all content. For a description of the topics, see Table 3 (Appendix). Not shown in the figure: Miscellaneous topic containing around 17% of posts.

The secondary keywords are useful for digging below the level of the 14 primary categories (Figure 10 in the Appendix). The most frequent keywords are generic, but the long tail picks up narrower themes that we did not anticipate when designing the taxonomy. For example, around 1.5% of messages mention “mental health” and around 2% contain a “call to action.” This kind of low-level tagging would have been very hard to obtain without a VLM in the loop.

Because we ran the same pipeline on text, images, and video, we can split topics by modality. Figure 3 shows the modality mix within each topic, and Figure 4 shows each topic-modality pair’s share of the overall corpus.

The within-topic view shows a lot of variation across topics. *Commerce* is dominated by images (around 60%), which

matches how the platform is used for product catalogs and flyers. *Greetings* and *Poetry* are mostly text.

The most relevant pattern for our paper is on the *Politics* side. About 65% of political content in our sample is non-textual (image or video). A text-only pipeline would miss nearly two-thirds of the political content in this corpus, which is one reason we think multimodal methods are necessary, not just useful, for studying WhatsApp.

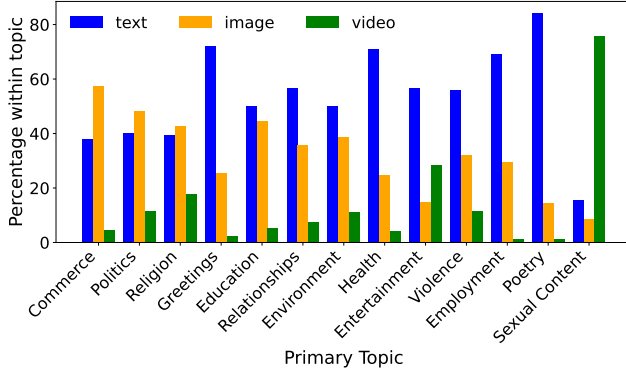


Figure 3: Modality percentage distribution within each primary topic.

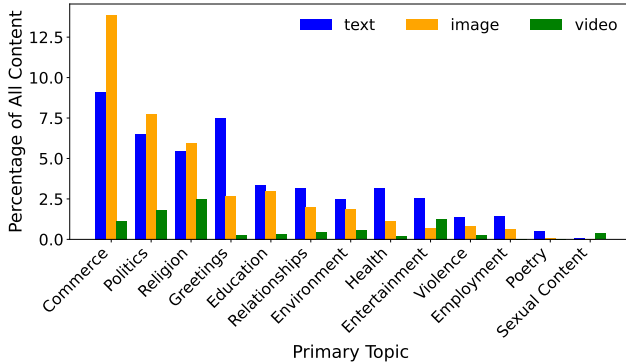


Figure 4: Modality share of total dataset per primary topic.

Finally, we look at virality by topic, using the share of messages tagged as “forwarded many times” within each category.

Figure 5 shows a clear split between high-arousal and utilitarian content. *Politics* and *Violence* have much higher viral rates than the overall baseline, while *Education*, *Greetings*, and *Employment* are almost entirely non-viral. The latter tend to be local, functional messages that stay within a single group; the former tend to be the kind of content that travels.

One surprising outlier is *Sexual Content*, with a viral rate of about 22%. It is not obvious why pornographic content would be forwarded across many hops on a personal messaging app. When we looked more closely, the high rate turned out to be driven by a small cluster of groups that exist mainly to circulate adult content.⁴

⁴We identified six groups responsible for a large share of sex-

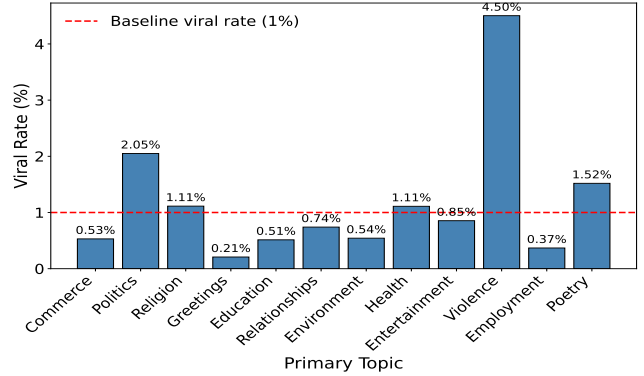


Figure 5: Virality composition (% viral vs. non-viral) for the primary topics.

Table 2 breaks this down further by modality. Some topic-modality pairs are much more viral than others: political and sexual videos, and both videos and images in the context of violence, stand out.

Primary Topic	Text	Image	Video
Commerce	0.16	0.71	1.60
Politics	1.10	1.43	10.32
Religion	0.33	0.86	2.81
Greetings	0.00	0.67	1.32
Education	0.33	0.45	2.55
Relationships	0.16	1.08	3.21
Environment	0.19	0.56	2.01
Health	0.75	1.19	8.87
Entertainment	0.03	0.61	2.15
Violence	0.75	5.67	24.32
Employment	0.30	0.57	0.00
Sexual Content	0.85	2.82	29.04
Poetry	1.08	1.82	60.00

Table 2: Viral Rate (%) by Topic and Modality.

5.2 Do LLMs Work for Multimodal Content Annotation

Our topic analysis shows that LLM/VLM representations are strong enough to organize WhatsApp content into broad, high-level themes. A natural next question is whether the same models can support *fine-grained* social annotation at scale, especially for labels that matter for governance and harm mitigation (e.g., misinformation, hate, and caste-based hostility). These labels are harder than topic tagging for two reasons: they are rare (so the training signal is weak), and they often depend on implicit meaning, local context, and contested norms rather than explicit keywords.

Figure 6 summarizes performance across nine binary tasks.⁵ Even with our best settings, results are mixed and,

ual video content, most of which is forwarded. These groups skew the overall ratio for the category and are not representative of the typical user.

⁵We report F1 because the class imbalance is severe and accuracy is uninformative.

for most tasks, not good enough to use without a human in the loop. The baseline classifiers do worst on the rarest categories. Imbalance ratios range from 2.2:1 for Political to 55:1 for Caste, and baseline F1 tracks the imbalance closely (Caste: 0.157; Health Advice: 0.271). Categories that are both common and easier to define on surface cues are easier: Political reaches a baseline F1 of 0.667.

To separate the role of class imbalance from the role of semantics, we also train versions of each classifier with SMOTE oversampling (Chawla et al. 2002). F1 improves for most labels, sometimes substantially. But the ceiling is still low for the harder categories: even after balancing, Caste only reaches 0.330, and several harm-oriented labels are still far from the precision required for moderation or high-stakes measurement. So imbalance is a big part of the story but not all of it. Some labels are hard because the positive class is defined by implication and social meaning (caste-coded hostility is the clearest example), not by stable surface cues.

A useful comparison is with the more explicit categories such as Political, Partisan, and News. We are not arguing that LLMs and VLMs “cannot” detect harmful content. We are arguing that, right now, they are much more dependable for coarse semantic labels (“is this political/news?”) than for socially grounded judgments about harm that need context and cultural reading (“is this hate/misinformation/casteist?”).

Performance also varies a lot by modality and by virality (Appendix Figure 12). Two points are worth noting. First, images and videos are not uniformly harder than text. In several tasks the visual modalities give comparable or slightly higher F1, which is somewhat surprising given how much visual content relies on cultural cues. Second, viral content is consistently easier to label than non-viral content. Highly forwarded items tend to be more explicitly framed and use more recognizable patterns. This matters as a measurement concern: if you only evaluate on viral content, the automated system will look better than it actually is on the broader background conversation.

We also checked whether prompt optimization could close the gap. We ran DSPy’s MIPROv2 prompt optimizer (Khatab et al. 2024) in prompt-only mode (no few-shot examples) on roughly 5,000 labeled examples per task, using Qwen2.5-7B-Instruct for both the task and the optimizer. Across the tasks we tried (political, caste-based, misinformation, hate), we saw no meaningful improvement, and in a few cases a small drop. The bottleneck is not the wording of the prompt. It is some combination of (i) heavy class imbalance, (ii) weak and noisy cues for many harm categories, and (iii) real ambiguity in the underlying labels.

5.3 Cross-Channel Narrative Overlap: Mainstream Media and WhatsApp

Our last analysis looks at how narratives from mainstream television show up on WhatsApp. We use the semantic retrieval pipeline from Section 4.3 to measure how much of the YouTube news content circulating during the election period is also visible in our WhatsApp data, including when the WhatsApp carrier is a video clip, screenshot, or meme rather than text. The goal here is not to argue that TV “causes” WhatsApp content or the other way around. The point is that

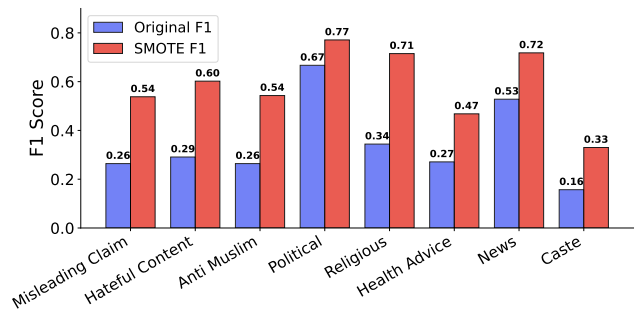


Figure 6: F1 performance of automated classifiers for nine content labels. Blue bars show baseline training under the natural class distribution; red bars show performance after training with SMOTE oversampling to mitigate class imbalance (Chawla et al. 2002).

this kind of cross-channel overlap can be measured at scale once images and videos are brought into the same analytical pipeline as text. We focus on two questions: (i) which mainstream narratives show up on WhatsApp during the 2024 election period, and (ii) which modalities carry them?

We start from the 98 YouTube topic clusters built in Section 4.3 and drop any topic with fewer than 10 matched WhatsApp messages, which leaves 45 topics. The rest of this section uses the 2,884 WhatsApp messages matched to these 45 topics.⁶

Even this conservative matched set shows that some mainstream television narratives reappear on WhatsApp recognizably, despite translation, remix, and modality changes. This is the kind of signal that a text-only matching pipeline tends to miss when the WhatsApp carrier is a screenshot or a clip rather than a transcript.

Of the 45 topics, 19 (42.2%) are pro-government, 6 (13.3%) are anti-government, and 20 (44.4%) are neutral/news (Appendix Table 4). The distribution of *messages* is more skewed than the distribution of *topics*: pro-government topics account for 58.7% of the matched WhatsApp messages. So mainstream narratives do not enter WhatsApp evenly; a few narrative types tend to reappear across many groups and are more likely to be repackaged into formats that travel well on the platform.

Two readings of this are consistent with the data. First, the YouTube corpus itself comes from a prime-time ecosystem in which government-aligned narratives are common, so the matched WhatsApp distribution mostly indicates that these narratives are also present in private messaging. Second, since the matched set is conditioned on what appears on television, this is best read as *coupling* between channels rather than one-directional influence. Below we show that this coupling is not just textual, and that the narratives that cross over most clearly are also disproportionately viral on WhatsApp.

Figure 7 shows the ten topics with the most matched WhatsApp messages. The top 4 by volume, and 6 of the

⁶The full topic list, our stance labels (pro-government, anti-government, neutral/news), and per-topic viral rates are in Appendix Table 4.

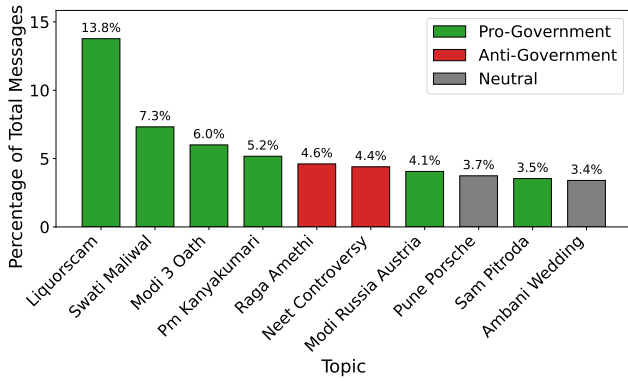


Figure 7: Volume of WhatsApp messages matching the top 10 mainstream news narratives, ranked by total message count.

top 10, are pro-government topics centered on promoting the Prime Minister or government programs. A small set of television narratives therefore accounts for a large share of cross-channel overlap, which is consistent with WhatsApp circulation tracking a relatively narrow set of high-salience frames during the election.

Because we match in a shared text representation that includes VLM descriptions of images and video, we can break the matched set down by modality. Figure 8 shows the modality mix for the top ten topics. Two things stand out. First, every high-volume narrative appears in more than one modality, so cross-channel transport is not purely textual. Second, the *dominant* carrier varies by narrative: some topics are carried mostly by videos (consistent with TV segments being forwarded as short clips), while others appear mostly as images (consistent with screenshots and memes). The same TV narrative tends to enter WhatsApp in whatever format preserves its framing best, which is often visual rather than textual.

Across all 45 topics, only 38.3% of matched messages are text. The remaining 61.7% is non-text, split between videos (32.3%) and images (29.4%). A text-only matching pipeline would therefore miss most of the measurable overlap between mainstream television narratives and WhatsApp circulation in our data.

We next ask whether cross-channel narratives are simply present on WhatsApp or whether they are also disproportionately viral. Across the 45 topics, the average share of messages tagged as viral is 28.35% (median 26.35%), far above the baseline of about 1% in the overall dataset (Figure 5). For the top ten topics, the viral share is even higher (36.71%). Even under a conservative reading, this gap is substantial: narratives strong enough to show up in both television coverage and identifiable WhatsApp content tend to be the kind of content that gets forwarded.

Viral rates also differ by stance. Pro-government topics have a higher average viral rate (41.44%) than neutral/news topics (19.70%) and anti-government topics (30.54%). The number of anti-government topics is small, so we are careful not to read these differences causally. But the direction is consistent with previous qualitative accounts of partisan am-

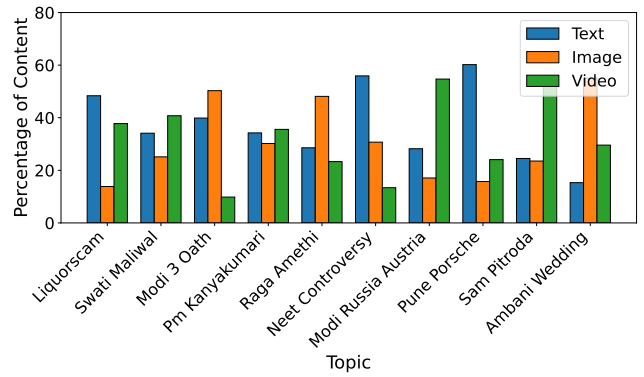


Figure 8: Modality distribution of matched WhatsApp messages for the top 10 topics. Each bar sums to 100% and shows the share of matched messages that are text, image, or video.

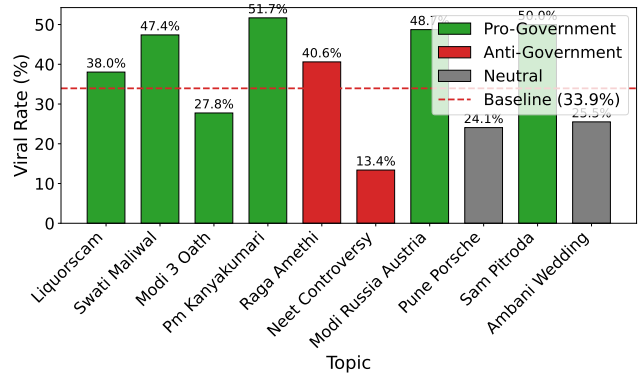


Figure 9: Viral versus non-viral composition of matched WhatsApp messages for the top 10 topics. Bars show counts, partitioned by WhatsApps “forwarded many times” label.

plification on WhatsApp, and it suggests that cross-channel coupling and WhatsApp virality may not be politically symmetric during elections.

Taken together, these results show that cross-channel coupling is measurable, that it is largely multimodal, and that it is concentrated in content that goes viral. Most of the overlap we detect would be missed by text-only matching, and the narratives that do overlap are disproportionately represented among viral WhatsApp messages. This is why we think multimodal LLM/VLM tooling is not just convenient but actually necessary if we want to connect WhatsApp to the broader media system around it.

6 Discussion

This paper has been about a practical measurement problem: how to study political communication on a platform whose content is multilingual and, in our setting, mostly images and videos. Across three analyses, treating WhatsApp as a multimodal medium changes what we can see, and current LLM/VLM tooling makes parts of this analysis tractable at a scale that would otherwise be hard to reach. At the same time, the models that are useful for exploration and coarse categorization are not reliable substitutes for human judg-

ment on culturally grounded categories such as misinformation, hate speech, and caste-coded hostility. Finally, multimodal representations make it possible to link private WhatsApp circulation to mainstream television narratives even when the WhatsApp carrier is a screenshot or a clip.

Why text-only analysis is not enough. Roughly 65% of political content in our data is non-textual (Garimella and Eckles 2020; Varanasi, Pal, and Vashistha 2022). Studies that rely only on scraped text or on public groups will undercount political content and will give a partial picture of what political mobilization on WhatsApp looks like, especially in settings like ours.

The reliance on visual formats (images for memes and posters, videos for news clips and speeches) is consistent with the “visual turn” in political communication (Veneti, Jackson, and Lilleker 2019). Where literacy is uneven and linguistic diversity is high, images and short clips travel more easily than long text. The VLM pipeline lets us go a bit beyond counting images and start describing what they actually do: Commerce content is visual because it is essentially a catalog, while Politics content is visual because it is emotive and symbolic. Being able to produce that kind of description without manual coding is what makes large-scale visual analysis of campaigning practical.

Models vs. local context. The pipeline does well on broad topics like Commerce and Politics but is considerably less reliable on categories that depend on social interpretation, local context, and contested definitions, including misinformation and hate speech. This is in line with previous work showing that LLM hate detection scores can look strong on benchmarks but remain sensitive to prompts, domain shift, and implicit context (Roy et al. 2023; Albladi et al. 2025). The failure modes we saw qualitatively are familiar in the Indian setting: coded language, humor, implicit targets, and cues that require cultural knowledge rather than surface semantics. Multimodal content makes this harder still, because the harmful content can be carried by composition, insinuation, or a visual trope rather than by an explicit slur.

One thing worth flagging from the annotation process is that disagreement is not only a model problem; it is also a social one. Our annotators were local to the region where the data were collected, and content that we (as authors) read as clearly hateful or clearly misleading was sometimes treated as ordinary or politically unsurprising by them. This does not mean there is no ground truth, but it does mean that labels like “hate speech” and “misinformation” are not purely technical. They depend on what a community has normalized, what it treats as credible, and what it considers harmful. Even with structured guidelines, these labels are hard to define in a way that is both locally meaningful and comparable across settings. The practical implication is not that automated labeling is hopeless. It is that model-based labels should be treated as measurements with error: useful for aggregate patterns and for hypothesis generation, but needing auditing, sensitivity checks, and human validation before being used for high-stakes claims.

The TV-WhatsApp boundary is porous. Our YouTube-to-WhatsApp matching is consistent with the picture of a hybrid media system in which broadcast and social channels

are closely linked (Garimella and Datta 2024). The ability to match multimodal artifacts (e.g. a TV news segment to a WhatsApp video clip) gives us a way to look at the “central source” hypothesis discussed in Indian media studies (Mahapatra and Plagemann 2019) with WhatsApp-side data. The fact that pro-government narratives are both more prevalent and more viral on WhatsApp is at least consistent with coordinated messaging between aligned television coverage and party-linked WhatsApp distribution, though we are careful not to read this as causal proof.

Our matched topic set is small, so we treat this part of the paper as primarily methodological. What we want to show is that it is technically possible to audit the gap between public broadcast and private messaging without relying on text matching alone or on direct video matching. That makes it easier to study how narratives are seeded by elites and then propagated through forwarding chains, which is otherwise mostly described through anecdotes about “IT cell” operations (Jakesch et al. 2021).

Limitations. There are several caveats. The WhatsApp dataset is collected through data donation and covers one state over a five-month window (UP, April–September 2024), so it is not a national sample. Voluntary donation likely over-represents users who are more digitally engaged or more willing to share, and the resulting sample may not capture the full range of WhatsApp use patterns. Our virality signal is WhatsApp’s “forwarded many times” tag, which tells us only that a message has been forwarded at least five hops; we do not observe full diffusion paths. Our model-based labels and matches can encode bias and can fail for culture-specific content; we audit failures qualitatively but do not treat these labels as ground truth. For the cross-channel analysis, YouTube uploads are not identical to the live broadcast and may be edited for the YouTube audience, and our topic extraction from video titles adds further measurement error. The whole pipeline runs on VLM-generated descriptions, so any systematic bias in how these models read Hindi or visual material specific to this context will propagate to topic modeling and matching.

Finally, the underlying models change quickly. The specific models we use (late 2024 / early 2025) will likely be replaced within a year, but the broader recipe of converting multimodal content into text descriptions and running unified text-based analysis on top should still apply.

Overall, the contribution of this paper is to show that multimodal LLM and VLM tooling can make large-scale WhatsApp analysis tractable, without claiming that the hardest interpretive tasks are solved. Bringing in images and video changes the empirical picture of what is discussed on WhatsApp; model-based annotation extends what we can measure but still needs auditing; and multimodal narrative matching gives us a new way to study how political communication moves between mainstream media and encrypted messaging.

Generative AI usage. The authors acknowledge the use of generative AI tools like Gemini for writing assistance in parts of the paper. All AI generated text was double checked by the authors before being included in the paper.

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Ethics checklist

1. For most authors...
 - (a) Would answering this research question advance science without violating social contracts, such as violating privacy norms, perpetuating unfair profiling, exacerbating the socio-economic divide, or implying disrespect to societies or cultures? **Yes**
 - (b) Do your main claims in the abstract and introduction accurately reflect the paper’s contributions and scope? **Yes**
 - (c) Do you clarify how the proposed methodological approach is appropriate for the claims made? **Yes**
 - (d) Do you clarify what are possible artifacts in the data used, given population-specific distributions? **Yes**
 - (e) Did you describe the limitations of your work? **Yes**
 - (f) Did you discuss any potential negative societal impacts of your work? **Yes**
 - (g) Did you discuss any potential misuse of your work? **No**
 - (h) Did you describe steps taken to prevent or mitigate potential negative outcomes of the research, such as data and model documentation, data anonymization, responsible release, access control, and the reproducibility of findings? **Yes**
 - (i) Have you read the ethics review guidelines and ensured that your paper conforms to them? **Yes**
2. Additionally, if your study involves hypotheses testing...
 - (a) Did you clearly state the assumptions underlying all theoretical results? **NA**
 - (b) Have you provided justifications for all theoretical results? **NA**
 - (c) Did you discuss competing hypotheses or theories that might challenge or complement your theoretical results? **NA**
 - (d) Have you considered alternative mechanisms or explanations that might account for the same outcomes observed in your study? **NA**
 - (e) Did you address potential biases or limitations in your theoretical framework? **NA**
 - (f) Have you related your theoretical results to the existing literature in social science? **Yes**
 - (g) Did you discuss the implications of your theoretical results for policy, practice, or further research in the social science domain? **Yes**
3. Additionally, if you are including theoretical proofs...
 - (a) Did you state the full set of assumptions of all theoretical results? **NA**
 - (b) Did you include complete proofs of all theoretical results? **NA**
4. Additionally, if you ran machine learning experiments...
 - (a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)? **No**
 - (b) Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)? **Yes**
 - (c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? **Yes**
 - (d) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? **NA**
 - (e) Do you justify how the proposed evaluation is sufficient and appropriate to the claims made? **Yes**
 - (f) Do you discuss what is “the cost” of misclassification and fault (in)tolerance? **No**
5. Additionally, if you are using existing assets (e.g., code, data, models) or curating/releasing new assets, **without compromising anonymity**...
 - (a) If your work uses existing assets, did you cite the creators? **Yes**
 - (b) Did you mention the license of the assets? **Yes**
 - (c) Did you include any new assets in the supplemental material or as a URL? **No**
 - (d) Did you discuss whether and how consent was obtained from people whose data you’re using/curating? **Yes**
 - (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? **Yes**
 - (f) If you are curating or releasing new datasets, did you discuss how you intend to make your datasets FAIR (see Wilkinson et al. (2016))? **Yes**
 - (g) If you are curating or releasing new datasets, did you create a Datasheet for the Dataset (see Gebru et al. (2021))? **NA**
6. Additionally, if you used crowdsourcing or conducted research with human subjects, **without compromising anonymity**...
 - (a) Did you include the full text of instructions given to participants and screenshots? **No**
 - (b) Did you describe any potential participant risks, with mentions of Institutional Review Board (IRB) approvals? **Yes**
 - (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? **Yes**
 - (d) Did you discuss how data is stored, shared, and de-identified? **Yes**

A Secondary keywords

Figure 10 shows the top 20 secondary keywords in our dataset.

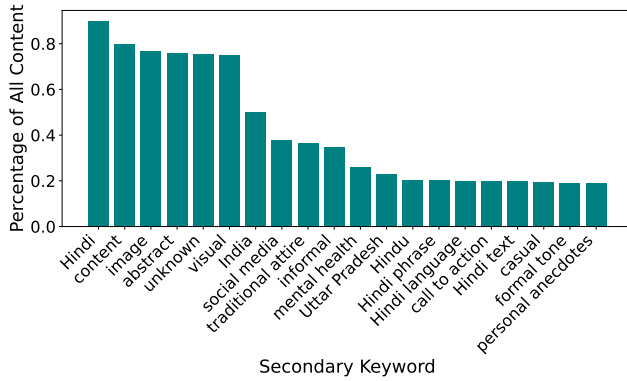


Figure 10: Top 20 secondary keywords as a percentage of all extracted keywords.

B Group coverage

To assess how content production is distributed across groups, we measured the number of distinct groups required to account for 90% of all messages within each major topic category. For every primary topic, groups were ranked by their message volume, and the cumulative distribution was computed to determine the minimal set achieving 90% coverage. The resulting counts, shown in Figure 11, provide a compact measure of topical concentration: smaller values indicate that discussion is dominated by a few highly active groups, whereas larger values imply a more diffuse participation pattern. As visible in the plot, topics such as *Commerce*, *Religion*, and *Politics* exhibit high group counts, reflecting broad and distributed engagement, while categories like *Employment* and *Poetry* reach 90% coverage with far fewer groups, suggesting more localized or niche conversation clusters.

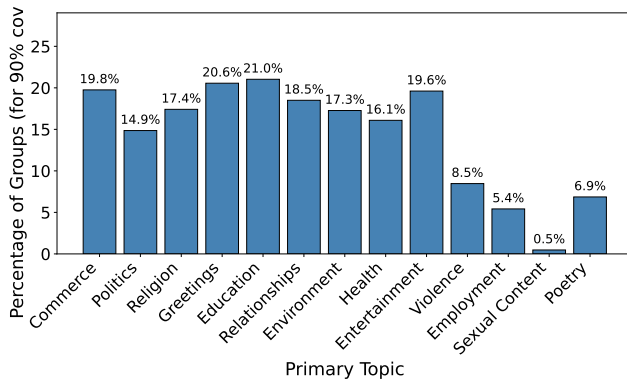


Figure 11: Number of distinct groups required to cover 90% of all messages for each primary topic. Lower values indicate higher topical concentration within fewer groups.

C Experiment Definitions

C.1 Caste Detection

Prompt:

Task: Determine whether this [Text/Image/Video] contains casteist elements.

Definition: Content is casteist if it makes any reference to caste explicitly or implicitly including stereotypes, discrimination, or social hierarchies based on caste.

Instructions:

1. Analyze the content thoroughly
2. Does this contain casteist content? (Yes/No)

Output Schema:

```
{
  "type": "object",
  "properties": {
    "caste": {"type": "boolean"}
  },
  "required": ["caste"]
}
```

C.2 Claim Detection

Prompt:

Task: Determine whether this [Text/Image/Video] contains a false or misleading claim.

Definition: Content is misleading if it misleads the audience. This includes: misrepresenting context, omitting important details, manipulative framing/editing, false information as fact.

Instructions:

1. Analyze all elements
2. Does it contain false/misleading claims? (Yes/No)

Output Schema:

```
{
  "type": "object",
  "properties": {
    "is_claim_misleading": {"type": "boolean"}
  },
  "required": ["is_claim_misleading"]
}
```

C.3 Hateful Content Detection

Prompt:

Task: Determine whether this [Text/Image/Video] contains hateful elements.

Definition: Content that promotes violence or incites hatred against individuals/groups based on: race, ethnic origin, religion, disability, gender, age, veteran status, or sexual orientation/gender identity.

Hateful content includes: dehumanizing speech or imagery, allegations of serious immorality/criminality, slurs or harmful stereotypes, serious insults/contempt/disgust, calls for exclusion or segregation.

Note: Criticism of political parties/leaders (BJP/Congress/Modi/Rahul Gandhi) or organizations (RSS) is not hateful.

Instructions:
1. Analyze completely
2. Is this hateful content? (Yes/No)
3. If Yes: Is it anti-Muslim? Does it align with common anti-Muslim narratives?

Output Schema:

```
{
  "type": "object",
  "properties": {
    "hateful_content": {"type": "boolean"},
    "anti_muslim": {"type": "boolean"},
    "common_theories": {
      "type": "string",
      "enum": [
        "Love Jihad",
        "Population Jihad",
        "Forced Conversions",
        "Muslim Appeasement",
        "None"
      ]
    }
  },
  "required": [
    "hateful_content",
    "anti_muslim",
    "common_theories"
  ]
}
```

C.4 Health Advice Detection

Prompt:

Task: Determine whether this [Text/Image/Video] contains health-related advice or claims.

Definition: Any content that makes claims linked to health, including: medical advice or treatments, health-related claims or remedies, disease prevention or cure suggestions, dietary or lifestyle recommendations, traditional medicine practices.

Instructions:
1. Analyze all content
2. Does this contain health-related content/advice? (Yes/No)

Output Schema:

```
{
  "type": "object",
  "properties": {
    "health_advice": {"type": "boolean"}
  },
  "required": ["health_advice"]
}
```

C.5 News Detection

Prompt:

Task: Determine whether this [Text/Image/Video] contains news-related information.

Definition: Content which shares: news clips or TV news footage, reporting

on current events, updates about incidents/developments, news-style presentations/formats, screenshots/clips from news sites/apps.

Instructions:
1. Analyze the content
2. Does this contain news-related content? (Yes/No)

Output Schema:

```
{
  "type": "object",
  "properties": {
    "news": {"type": "boolean"}
  },
  "required": ["news"]
}
```

C.6 Political Content Detection

Prompt:

Task: Determine whether this [Text/Image/Video] is political and analyze partisan characteristics.

Definition: Political content that: refers to political parties or leaders, discusses politically relevant issues, shows political rallies/speeches/campaigns, can be explicit or implicit.

Partisan Content: Political content taking a party/ideology side.

- Non-partisan: Content discussing pros/cons of a

law without party refs

- Partisan: Election results with captions praising specific parties

Propaganda Classification (if partisan): BJP (praising Modi/Yogi/Amit Shah, BJP rallies), Congress (praising Rahul Gandhi, Congress events), Other (Hindutva-aligned without explicit party).

Instructions:
1. Analyze all aspects
2. Is this political? (Yes/No)
3. If Yes: Is it partisan? (Yes/No)
If partisan, which party/group? (BJP/Congress/Other)

Output Schema:

```
{
  "type": "object",
  "properties": {
    "political": {"type": "boolean"},
    "partisan": {"type": "boolean"},
    "propaganda_for": {
      "type": "string",
      "enum": ["BJP", "Congress", "Other"]
    }
  },
  "required": [
    "political",
    "partisan",
    "propaganda_for"
  ]
}
```

C.7 Religious Content Detection

Prompt:

Task: Determine whether this [Text/Image/Video] contains religious elements and iconography.

Definition: Content relating to: religious groups, religious beliefs, religious images/symbols/ceremonies, explicit or implicit references.

Religious Iconography includes: images/videos of gods and goddesses, religious symbols displayed or worn, religious stories or narratives, religious ceremonies or rituals, religious buildings or places of worship.

Instructions:

1. Analyze all content
2. Does this contain religious content? (Yes/No)
3. If Yes: Does it contain religious iconography? (Yes/No)

Output Schema:

```
{
  "type": "object",
  "properties": {
    "religious": {"type": "boolean"},
    "religious.iconography": {"type": "boolean"}
  },
  "required": [
    "religious",
    "religious.iconography"
  ]
}
```

C.8 Misinformation Detection

Prompt:

Task: Determine whether this [Text/Image/Video] contains misinformation and assess verifiability.

Definition: False information spread through content, regardless of intent to mislead.

Classification Levels: Yes (definitely contains misinformation), Maybe (potentially contains misinformation), No (does not contain misinformation).

Verifiability: For Yes/Maybe, determine if misinformation is theoretically verifiable. Theoretically verifiable: A single investigative journalist could fact-check the claim. Unverifiable: Cannot be definitively proven/disproven (e.g., "Group X always does Y").

Instructions:

1. Analyze all content
2. Does this contain misinformation? (Yes/Maybe/No)
3. If Yes or Maybe: Can it be theoretically verified?

Output Schema:

```
{
  "type": "object",
  "properties": {
```

```
    "misinformation.part": {
      "type": "string",
      "enum": ["yes", "no", "maybe"]
    },
    "can.theoretically.verify": {
      "type": "boolean"
    }
  },
  "required": [
    "misinformation.part",
    "can.theoretically.verify"
  ]
}
```

C.9 Content Description

Prompt:

Please provide a detailed description of this [Text/Image/Video].

Instructions:

1. Analyze thoroughly including: visual content and scenes, audio content (speech, music, sounds), text overlays or captions, overall narrative or message.
2. Provide comprehensive description capturing: main topic/subject matter, key points/arguments/story, tone and style, notable features/editing/production elements.
3. Keep description clear and concise (3-4 sentences).

Output Schema:

```
{
  "type": "object",
  "properties": {
    "description": {"type": "string"}
  },
  "required": ["description"]
}
```

C.10 Topic Modeling

Prompt:

Task: Identify the primary topic and five descriptive keywords for the given text, image, or video.

Primary Topic: Choose ONE from the following list of 14 topics: Religion, Politics, Entertainment, Commerce, Poetry, Employment, Violence, Health, Education, Environment, Greetings, Relationships, Sexual Content, Miscellaneous

Secondary Keywords: Extract a list of keywords describing the content.

Instructions:

1. Select exactly one primary topic.
2. Return a JSON array of keyword phrases.
3. Select between 5 and 10 keywords.
4. Keywords should be neither too specific nor too generic.

Output Schema:

```
{
  "type": "object",
  "properties": {
    "primary_topic": {
      "type": "string",
      "enum": [
        "Religion", "Politics", "Entertainment",
        "Commerce", "Poetry", "Employment",
        "Violence", "Health", "Education",
        "Environment", "Greetings",
        "Relationships", "Sexual Content",
        "Miscellaneous"
      ]
    },
    "secondary_keywords": {
      "type": "array",
      "items": {"type": "string"},
      "minItems": 5,
      "maxItems": 10
    },
    "required": [
      "primary_topic",
      "secondary_keywords"
    ]
  }
}
```

Message 1: Commerce (Non-viral, 09/03/2024 02:14)

Original Text (Hindi): 12 July ko Anant [ENCRYPTED] Mukesh Ambani ji de rahe hain pure Bharat ko free mein Rs.749 wala 3 m[ENCRYPTED] link par click karke apne number par recharge kare. <https://tirangagame.today/>

Description: This text is a promotional message for a free recharge offer by Mukesh Ambani. It is written in Hindi and includes an encrypted link. The message is promotional in nature and uses a casual tone.

Topic: Commerce **Keywords:** recharge offer, Mukesh Ambani, promotional message, Hindi language, encrypted link

Detection Category	Result
Caste	False
Claim	Misleading: False
Hateful	Content: False, Anti-Muslim: False, Theories: None
Health	Advice: False
News	False
Political	Political: False, Partisan: False, Propaganda: Other
Religious	Iconography: False, Religious: False
Misinformation	Present: Maybe, Verifiable: False
SCAW	Present: False, Verifiable: False, Scope: National

D Examples from Dataset

D.1 Text

Message 2: Politics (Viral, 16/07/2024 02:43)

Original Text (Hindi): Subject: Statement on the extreme oppression of General Category people. Honorable Prime Minister and nation's [ENCRYPTED] leader, please emphasize this system so that General Category people are not oppressed... [Long message comparing experiences of General Category vs Reserved Category students across entrance exams, college fees ([ENCRYPTED]), mess fees ([ENCRYPTED]), pocket money ([ENCRYPTED]), CAT scores, campus recruitment, and GATE scores. Ends with call to forward message to save country from "reservation termites"]

Description: The text is a letter or statement written in Hindi, addressing the issue of discrimination against General Category students in India. It starts with a formal salutation to the Prime Minister, narrates comparative experiences of a General Category student and a reserved category student highlighting disparities in academic and professional achievements. The tone is passionate and emotional, with a strong call to action urging readers to forward the message.

Topic: Politics **Keywords:** discrimination, General Category students, Prime Minister, call to action, academic achievements, equal treatment

Detection Category	Result
Caste	False
Claim	Misleading: False
Hateful	Content: False, Anti-Muslim: False, Theories: None
Health	Advice: False
News	False
Political	Political: True, Partisan: True, Propaganda: BJP
Religious	Iconography: False, Religious: False
Misinformation	Present: Yes, Verifiable: True
SCAW	Present: True, Verifiable: True, Scope: National

Message 3: Politics (Non-viral, 06/05/2024 12:58)

Original Text (Hindi): “..Main apne Bahujan Samaj ke logon ko kahunga ki Bahujan Samaj Party ko vote dena apna vote kharab karna hai, wo BJP ka saath de rahe hain chahe saamne se chahe peeche se..” Akhilesh Yadav ka Mayawati par seedha hamla. Jaunpur aur Basti ke BSP ke ticket achanak badle gaye to Akhilesh Yadav ne BSP par direct attack kar diya. Jaunpur se Dhananjay Singh ki patni Shri Kala Singh aur Basti se Daya Shankar Mishra ka ticket badal kar dusre p[ENCRYPTED] nomination ka aakhiri din tha.

Description: The text discusses political maneuvering in India, specifically mentioning Akhilesh Yadav's direct attack on Mayawati and BSP in the context of the Uttar Pradesh elections. It highlights the sudden change in BSP tickets for candidates in Jaunpur and Basti, and mentions it was the last day of nomination.

Topic: Politics **Keywords:** elections, political maneuvering, Uttar Pradesh, Akhilesh Yadav, BJP, Samajwadi Party, BSP, campaigning

Detection Category	Result
Caste	False
Claim	Misleading: False
Hateful	Content: False, Anti-Muslim: False, Theories: None
Health	Advice: False
News	True
Political	Political: True, Partisan: True, Propaganda: Congress
Religious	Iconography: False, Religious: False
Misinformation	Present: No, Verifiable: False
SCAW	Present: True, Verifiable: True, Scope: National

Message 4: Miscellaneous (Non-viral, 17/09/2023 06:19)

Original Text (Hindi): Shilp vidya ke pravartak hone ke karan Bhagwan Vishwakarma ko nirman aur srijan ka aradhya mana jata hai. [ENCRYPTED] ko khubsurti di. Kisi angadh vastu ko akarshak banane ka kaam shilpkar karta hai. Is tarah ke karyon mein pratyakshat [ENCRYPTED] hai jaisa mastishk nirdesh deta hai. Manushya sharir mein sarvadhik mahatva mastishk ka hi hai. Isliye vidwanon ne sahitya, sangeet aur kala se anabhigya vyakti ko pashu saman kaha hai... Jai Shri Sitaram, Jai Vishwakarma Bhagwan [ENCRYPTED]

Description: The text discusses the importance of craftsmanship and creativity, associating it with the Hindu god Vishwakarma, the architect of the gods. It emphasizes the role of the mind in creating appealing objects and compares the knowledgeable person to a fragrant flower spreading the scent of knowledge. The tone is reverent and educational, concluding with religious invocations.

Topic: Miscellaneous **Keywords:** Vishwakarma, craftsmanship, creativity, knowledge, Hindu deity

Detection Category	Cate-	Result
Caste		False
Claim		Misleading: False
Hateful		Content: False, Anti-Muslim: False, Theories: None
Health		Advice: False
News		False
Political		Political: False, Partisan: False, Propaganda: Other
Religious		Iconography: True, Religious: True
Misinformation		Present: No, Verifiable: False
SCAW		Present: False, Verifiable: False, Scope: Local

Message 5: Commerce (Non-viral, 05/07/2024 13:56)		
Original Text (Hindi): Lucknow - Rajdhani mein Nagar Nigam bana raha hai Mango Park. Raibareli Road par Kisan Path ke paas hoga nirman. Lagbhag 18 crore ki lagat se banega Mango Park. Karib 15 acre mein viksit kiya ja raha Mango Park. 108 prajatiyon ke 2068 paudhon ka ropan kiya jayega. 400 square meter mein Mango Museum banane ki योजना. Description: The text describes the development of a mango park in Lucknow, India. The park will be located near the farmer's path on Raibareli Road, and is expected to cost approximately 18 crore rupees. It will cover about 15 acres and will feature 2068 mango trees of 108 varieties. Additionally, a 400 square meter mango museum is planned. Topic: Commerce Keywords: mango park, Lucknow, India, mango trees, mango museum, 15 acres, 108 varieties, 18 crore rupees		
Detection Category	Cate-	Result
Caste		False
Claim		Misleading: False
Hateful		Content: False, Anti-Muslim: False, Theories: None
Health		Advice: False
News		False
Political		Political: False, Partisan: False, Propaganda: Other
Religious		Iconography: False, Religious: False
Misinformation		Present: No, Verifiable: True
SCAW		Present: False, Verifiable: False, Scope: Local

D.2 Images

Image 1: Commerce (Non-viral, 16/01/2024 05:46)	
 Description: The image displays a stack of women's kurtas in various pastel colors – beige, pink, green, and red – with intricate embroidered designs. Each kurta has price tags indicating they cost Rs. 567 each, while one tag on top shows Rs. 907. The overall presentation suggests this is likely an advertisement for clothing retail, showcasing affordable fashion options. Topic: Commerce Keywords: women's clothing, kurtas, pastel colors, intricate embroidery, price tags, retail advertisement, affordable fashion	
Detection Category	Result
Caste	False
Claim	Misleading: False
Hateful	Content: False, Anti-Muslim: False, Theories: None
Health	Advice: False
News	False
Political	Political: False, Partisan: False, Propaganda: Other
Religious	Iconography: False, Religious: False
Misinformation	Present: No, Verifiable: False
SCAW	Present: True, Verifiable: True, Scope: Local

Image 2: Politics (Non-viral, 04/06/2024 03:32)	
 Description: The image is a live election results screen from an Indian news channel, likely covering the 2024 elections. It displays seat counts for different political alliances – NDA (National Democratic Alliance), INDIA (a coalition of opposition parties), and OTH (others) – across various states like Uttar Pradesh, Maharashtra, and Bihar. The visual layout emphasizes numerical data with bold colors to highlight each alliance's performance in specific regions alongside images of politicians involved. Topic: Politics Keywords: election results, Indian	

news channel, political alliances, 2024 elections, seat counts

Detection Category	Result
Caste	False
Claim	Misleading: False
Hateful	Content: False, Anti-Muslim: False, Theories: None
Health	Advice: False
News	True
Political	Political: True, Partisan: True, Propaganda: BJP
Religious	Iconography: False, Religious: False
Misinformation	Present: Yes, Verifiable: True
SCAW	Present: True, Verifiable: True, Scope: National

Detection Category	Result
Caste	False
Claim	Misleading: True
Hateful	Content: False, Anti-Muslim: False, Theories: None
Health	Advice: False
News	True
Political	Political: True, Partisan: True, Propaganda: Other
Religious	Iconography: False, Religious: True
Misinformation	Present: Maybe, Verifiable: True
SCAW	Present: True, Verifiable: True, Scope: National

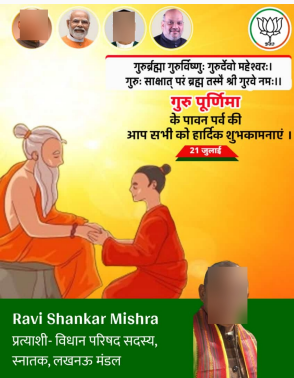
Image 3: Politics (Non-viral, 20/08/2024 14:38)



Description: The image shows a scanned newspaper clipping in Hindi discussing a Supreme Court order dated 23/06/2023 regarding land disputes, specifically concerning Wakf Boards. The article details how the court ruled against recognizing post-partition claims on Wakf properties by Bangladesh (formerly East Pakistan), stating these claims are not valid as they lack proper documentation; it also touches upon historical context dating back to 1947 and emphasizes government authority over such lands.

Topic: Politics **Keywords:** Supreme Court, land disputes, Wakf Boards, post-partition claims, historical context

Image 4: Politics (Non-viral, 21/07/2024 04:58)



Description: This is a political greeting card for Guru Purnima, featuring Indian Prime Minister Narendra Modi alongside BJP party symbols and imagery related to Hindu spirituality. The central image depicts two figures in prayer with text wishing recipients well on July 21st, while at the bottom it includes Ravi Shankar Mishra's name and his position as a candidate for Legislative Council member from Lucknow District. The overall tone is respectful and celebratory, aiming to connect religious observance with political messaging.

Topic: Politics **Keywords:** greeting card, Guru Purnima, Indian Prime Minister, BJP party, Hindu spirituality, political messaging, Legislative Council, candidate, Lucknow District, religious observance

Detection Category	Result
Caste	False
Claim	Misleading: False
Hateful	Content: False, Anti-Muslim: False, Theories: None
Health	Advice: False
News	False
Political	Political: True, Partisan: True, Propaganda: BJP
Religious	Iconography: True, Religious: True
Misinformation	Present: No, Verifiable: False
SCAW	Present: True, Verifiable: True, Scope: National

Detection Category	Result
Caste	False
Claim	Misleading: True
Hateful	Content: False, Anti-Muslim: False, Theories: None
Health	Advice: False
News	True
Political	Political: True, Partisan: False, Propaganda: Other
Religious	Iconography: False, Religious: False
Misinformation	Present: Maybe, Verifiable: True
SCAW	Present: True, Verifiable: True, Scope: Local

D.3 Videos

Image 5: Violence (Non-viral, 19/12/2023 05:08)



Description: The image depicts a tense standoff between police officers and a crowd of people, likely during a protest or public disturbance in an urban setting. Several uniformed policemen are attempting to control individuals within the densely packed group, with visible expressions of frustration and resistance on both sides; the scene is dimly lit suggesting it is nighttime or indoors, contributing to a somber and potentially volatile atmosphere.

Topic: Violence **Keywords:** police, crowd, protest, control, resistance, urban, tension, nighttime

Video 1: Religion (Viral, 06/05/2024 11:34)

Video Link: <http://bit.ly/3LKaS62>

Description: The video shows a religious procession in front of a temple. A woman in a pink and green sari performs a classical Indian dance in front of a large group of people, mostly women, who are singing and carrying small oil lamps. The group is singing a devotional song in Hindi, and the dancer's movements are synchronized with the music. The temple is decorated with lights, and there is a colorful rangoli on the ground. The procession ends with a decorated cart carrying statues of deities.

Topic: Religion **Keywords:** religious procession, classical Indian dance, devotional song, Hindi, temple decoration

Detection Category	Result
Caste	False
Claim	Misleading: False
Hateful	Content: False, Anti-Muslim: False, Theories: None
Health	Advice: False
News	False
Political	Political: False, Partisan: False, Propaganda: Other
Religious	Iconography: True, Religious: True
Misinformation	Present: No, Verifiable: False
SCAW	Present: True, Verifiable: True, Scope: Local

Video 2: Politics (Viral, 10/05/2024 16:26)

Video Link: <https://bit.ly/4bH7Xp8>

Description: The video is a political commentary criticizing the current Indian government led by Prime Minister Narendra Modi. It features a woman speaking directly to the camera, interspersed with black and white clips of Modi speaking, as well as images and video clips highlighting various issues. The woman argues that Modi's policies have led to increased unemployment, rising fuel prices, high gas cylinder costs, a large national debt, increased poverty, and the favoring of his friends with the country's wealth. The video concludes by urging viewers not to give Modi another chance in power.

Topic: Miscellaneous **Keywords:** political commentary, Modi criticism, unemployment, inflation, farmers protests

Detection	Cate-	Result
gory		
Caste		False
Claim		Misleading: True
Hateful		Content: False, Anti-Muslim: False, Theories: None
Health		Advice: True
News		True
Political		Political: True, Partisan: True, Propaganda: Other
Religious		Iconography: True, Religious: True
Misinformation		Present: Maybe, Verifiable: True
SCAW		Present: True, Verifiable: True, Scope: National

Video 3: Miscellaneous (Viral, 01/03/2024 17:37)

Video Link: <https://bit.ly/4pF0bzC>

Description: The video begins with a man in a blue shirt sitting across from a jeweler, who is calculating the cost of jewelry the man is selling. The jeweler tells him the total is 355,000, but the man says he is still 150,000 short. The man explains that his daughter is sick and needs an operation costing 300,000 to 500,000. The jeweler remembers UNM Children's Hospital in Kamrej that offers subsidized or free operations for children. The video shows the family visiting the hospital, meeting Dr. Rajesh Shah and Dr. Manish Jain, who assure them the operation can be done at subsidized rates.

Topic: Miscellaneous **Keywords:** hospital promotion, subsidized healthcare, children's surgery, family struggle, medical costs

Detection	Cate-	Result
gory		
Caste		False
Claim		Misleading: False
Hateful		Content: False, Anti-Muslim: False, Theories: None
Health		Advice: True
News		True
Political		Political: True, Partisan: True, Propaganda: BJP
Religious		Iconography: True, Religious: True
Misinformation		Present: Maybe, Verifiable: True
SCAW		Present: True, Verifiable: True, Scope: Local

Video 4: Politics (Non-viral, 06/04/2024 10:22)

Video Link: <https://bit.ly/4qVxrn5>

Description: The video depicts large-scale protests in Israel against Prime Minister Benjamin Netanyahu’s government over its handling of recent events involving war and hostage negotiations. The first scene shows an aerial view of massive crowds gathered on city streets at night under bright streetlights. Subsequent frames capture close-up shots of protesters holding signs with messages like “FREE ISRAEL FROM NETANYAHU” and images depicting political figures. Protesters express frustration through slogans such as “This government is doing horrendous things.”

Topic: Politics **Keywords:** protests, government, Netanyahu, dissent, accountability

Detection	Cate-	Result
gory		
Caste		False
Claim		Misleading: True
Hateful		Content: False, Anti-Muslim: False, Theories: None
Health		Advice: False
News		True
Political		Political: True, Partisan: False, Propaganda: Other
Religious		Iconography: False, Religious: False
Misinformation		Present: Maybe, Verifiable: True
SCAW		Present: True, Verifiable: True, Scope: National

Video 5: Environment (Non-viral, 04/07/2024 15:34)

Video Link: <https://bit.ly/4sHmIOX>

Description: The video shows close-up footage of someone wearing orange gloves gently handling a small white bird with red markings on its head and wings. The person appears to be examining or caring for the bird carefully while holding it in their hands. In the background, there is an outdoor setting featuring potted plants with green leaves, suggesting a garden-like environment. There is no visible text overlay or audio present.

Topic: Environment **Keywords:** bird care, garden setting, orange gloves, red markings, outdoor environment

Detection	Cate-	Result
gory		
Caste		False
Claim		Misleading: True
Hateful		Content: False, Anti-Muslim: False, Theories: None
Health		Advice: False
News		False
Political		Political: False, Partisan: False, Propaganda: Other
Religious		Iconography: False, Religious: False
Misinformation		Present: No, Verifiable: True
SCAW		Present: False, Verifiable: True, Scope: National

E Topic Explanations

Table 3 contains the primary topic explanations.

F Performance Tables

Figure 12 shows performance of different experiments across different modalities and viralities and the SMOTE vs baseline performance.

G Topics Identified from News and Used for Matching

Table 4 has the list of topics analyzed and information regarding the matched messages.

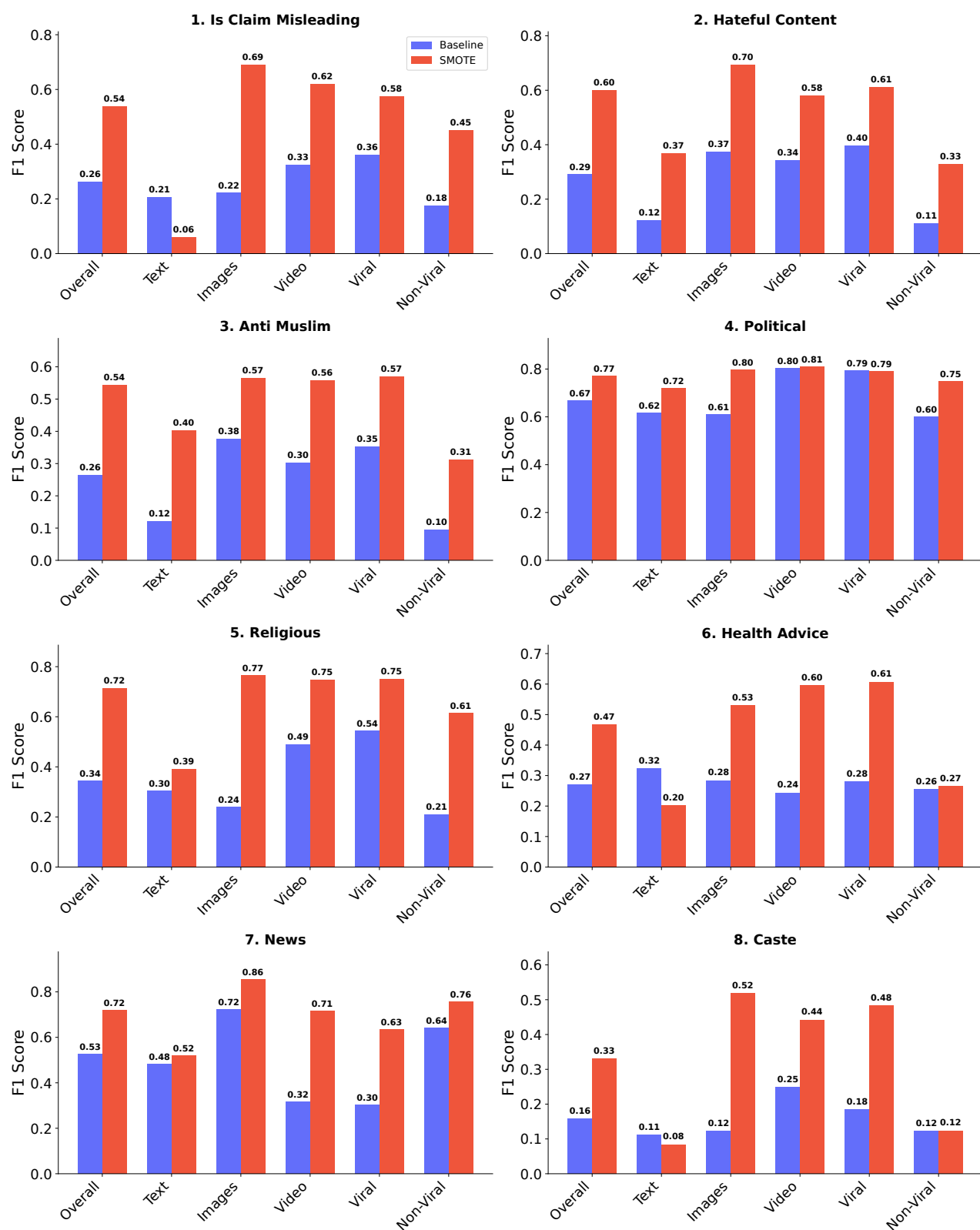


Figure 12: Performance of the large language models on various categories like Misinformation, Hate speech and topics like Politics, religion, etc.

Topic	Definition
Commerce	Messages related to buying, selling, pricing, promotions, or commercial transactions.
Politics	Discussions of political actors, parties, elections, public policy, or governance.
Religion	Content referencing religious beliefs, practices, institutions, or identities.
Greetings	Short social interactions such as greetings, wishes, congratulations, or casual salutations.
Education	Messages concerning schools, exams, learning, teaching, or educational institutions.
Entertainment	Content related to movies, music, celebrities, sports, or leisure activities.
Environment	Discussions about nature, climate, pollution, conservation, or environmental issues.
Health	Messages about physical or mental health, medicine, diseases, or healthcare services.
Violence	References to physical harm, threats, conflict, or violent acts.
Employment	Content related to jobs, recruitment, work conditions, or career opportunities.
Poverty	Discussions of economic hardship, inequality, unemployment, or lack of basic resources.
Sexual Content	Messages containing sexual references, explicit language, or adult themes.

Table 3: Primary topic categories with one-line definitions used for chat message classification.

#	Topic	Description	Stance	Msgs	Viral %
1	Liquorscam	Delhi liquor policy controversy	Pro-Gov	397	38.04
2	Swati Maliwal	AAP MP assault allegations	Pro-Gov	211	47.39
3	Modi 3 Oath	PM Modi third term swearing-in	Pro-Gov	173	27.75
4	Pm Kanyakumari	PM meditation at Vivekananda Rock	Pro-Gov	149	51.68
5	Raga Amethi	Rahul Gandhi Amethi election	Anti-Gov	133	40.60
6	NEET Controversy	Medical entrance exam irregularities	Anti-Gov	127	13.39
7	Modi Russia Austria	PM foreign visits to Russia and Austria	Pro-Gov	117	48.72
8	Pune Porsche	Teen drunk driving accident	Neutral	108	24.07
9	Sam Pitroda	Congress leader controversial statements	Anti-Gov	102	50.00
10	Ambani Wedding	Celebrity wedding of Mukesh Ambani's son Anant	Neutral	98	25.51
11	Kejriwal Release	Delhi CM Arvind Kejriwal release from custody	Anti-Gov	97	37.11
12	Ram Navmi	Hindu festival celebrations	Neutral	97	16.49
13	India T20 Win	Cricket World Cup victory	Neutral	95	4.21
14	Muslim Reservation	Debate on reservation policies	Pro-Gov	93	53.76
15	Union Budget	2024 Union Budget announcements	Pro-Gov	90	26.67
16	PM Modi Campaigns	PM election rallies and speeches	Pro-Gov	85	60.00
17	Delhi Water	Water crisis in national capital	Neutral	65	32.31
18	G7	PM Modi at G7 summit	Pro-Gov	58	39.66
19	Padma Awards	Topic related to Padma Awards	Neutral	45	8.89
20	Hathras Incident	Stampede at religious gathering	Anti-Gov	42	19.05
21	Heatwave	Extreme heat conditions across India	Neutral	42	35.71
22	OBC Reservation	OBC reservation debates	Pro-Gov	42	47.62
23	Babu Swearing In	Topic related to babu swearing in	Neutral	34	35.29
24	Kargil Diwas	Kargil War commemoration	Pro-Gov	29	17.24
25	Owaisi Jai Palestine	AIMIM leader pro-Palestine statement	Pro-Gov	25	48.00
26	Kanwar Yatra	Hindu pilgrimage and related controversies	Neutral	24	8.33
27	Ebrahim Raisi Death	Iranian President helicopter crash	Neutral	24	12.50
28	Delhi Coaching Flood	Topic related to Delhi coaching flood	Neutral	22	22.73
29	Naveen Swearing In	Topic related to Naveen swearing in	Neutral	22	18.18
30	Intl Yoga Day	International Yoga Day celebrations	Neutral	21	9.52
31	Amit Shah Fake	Controversy over fake video of Home Minister	Pro-Gov	19	57.89
32	Arvind Kejriwal Jail	Delhi CM arrest and legal proceedings	Pro-Gov	19	42.11
33	Raaj Anand	Political figure developments	Neutral	18	27.78
34	Gourav Vallabh	Congress leader joining BJP	Pro-Gov	17	35.29
35	Ram Sacrificial Goat	Religious controversy	Neutral	16	31.25
36	Airport Roof Collapse	Infrastructure failure at Delhi Airport	Anti-Gov	15	40.00
37	Evm Manipulation	Topic related to EVM manipulation	Neutral	14	21.43
38	Trump Shot	Topic related to Trump being shot	Neutral	13	0.00
39	Train Derailed	Train accident in Assam	Neutral	13	0.00
40	Jagan Mohan	Andhra Pradesh political developments	Neutral	13	23.08
41	Assam Campaign	Assam CM Himanta Biswa election campaigning	Pro-Gov	13	7.69
42	Keshav Maurya	Topic related to Keshav Maurya	Neutral	13	15.38
43	Kangana Slap	BJP MP slapped at airport	Pro-Gov	12	16.67
44	Prajwal Revanna	Karnataka politician controversy	Pro-Gov	12	16.67
45	Covishield Sideeffects	COVID vaccine side effects controversy	Anti-Gov	10	20.00

Table 4: Complete list of topics analyzed with descriptions and political stance classification.